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REVISED PALEOMAGNETIC POLE FOR THE SONOMA VOLCANICS, CALIFORNIA

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Abstract. Paleomagnetic sampling of the Miocene and Pliocene Sonoma Volcanics, northern California, was undertaken to supplement an earlier collection. Data from 25 cooling units yield positive fold and reversal tests, and a paleomagnetic pole located at 80.2°N., 069 2°E., with α_{95} = 6.8°. This paleopole is significantly displaced (9.6° ± 5.3° of latitude) to the farside of the geographic pole. A highly elliptical distribution of the data in both direction and VGP space indicates that incomplete averaging of geomagnetic secular variation is a more likely explanation for this anomaly than is northward translation of the volcanic field.

Introduction

The Sonoma Volcanics is part of a Neogene volcanic field in the California Coast Ranges north of the San Francisco Bay area (Figure 1). The volcanic rocks range in composition from basalt to rhyolite and include significant volumes of welded and unwelded ash-flow tuff. Detailed geologic maps are provided by Fox et al. [1973] and Sims et al. [1973]. Potassium-argon determinations by Evernden and James [1964], Mankinen [1972], Sarna-Wojcicki [1976], and Fox et al. [1985*b*] indicate a Miocene and Pliocene age for the formation. Although most ages range from approximately 8 Ma to 3 Ma, older volcanic rocks (14 Ma to 8.5 Ma) also are present [Mankinen, 1972; Fox et al., 1985*b*] and probably are part of the Tolay Volcanics of Morse and Bailey [1935].

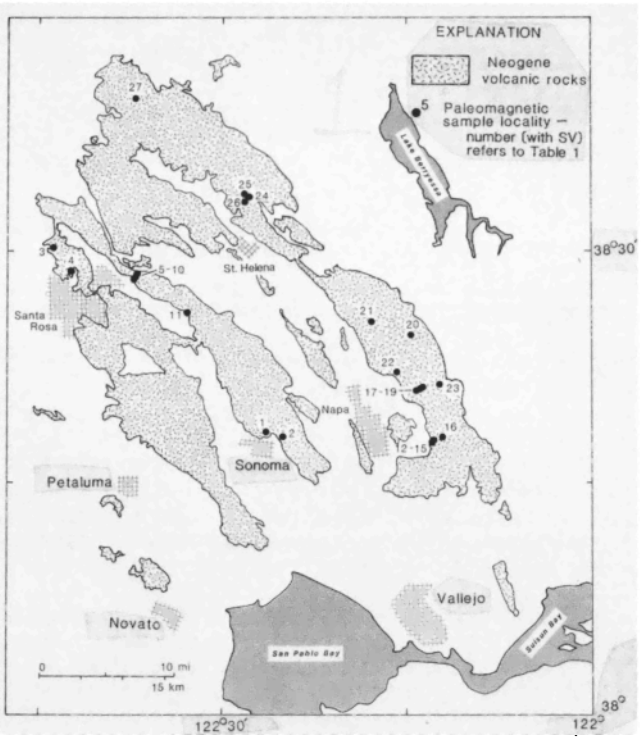


Fig. 1. Sketch showing major occurrences of the Sonoma Volcanics and paleomagnetic sample localities (Table 1). Modified from Koenig [1963].

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Paleomagnetic data from the Sonoma Volcanics yielded a mean paleopole that differed from the Earth's rotational axis [Mankinen, 1972]; a discordance that could be interpreted as indicating a significant amount (9.7° ± 6.9°) of northward translation relative to the stable continental interior. A paleolatitude anomaly of this magnitude was unexpected for rocks as young as these, and Beck [1980] commented that it was an "embarrassing" consequence of his selection criteria that the Sonoma Volcanics paleopole was identified as discordant. Although Fox et al. [1985*a*] did infer some northward translation for parts of the Sonoma Volcanics on the basis of offsets along major strike-slip faults, the amount was much smaller (30 to 70 km) than that indicated by the paleomagnetic data (more than 300 km). More than 300 km of lateral motion requires a displacement rate of more than 6 cm/yr, which is the estimated rate of relative motion between the Pacific and North American plates during the past 5 to 10 million years [Minster and Jordan, 1978]. Moreover, the volcanic field is northeast of the San Andreas fault zone, along which most of Minster and Jordan's relative motion occurred. A more likely explanation of the paleolatitude anomaly, therefore, is that the number of flows used to calculate the Sonoma Volcanics paleopole (14) was too small to ensure complete averaging of paleosecular variation. On the other hand, Beck [1984] noted that most lower and middle Tertiary rocks within the Washington-Oregon Coast Range had paleomagnetic inclinations that were shallower than expected for their present latitude. Although the discordance seldom is significant at the 95% confidence level, the bias toward shallow inclinations led Beck to speculate that those rocks moved about 350 km northward as a result of oblique subduction along the continental margin. Additional sampling of the Sonoma Volcanics, therefore, seemed warranted in order to determine a more accurate paleopole for the formation and to evaluate further the possibility of significant and very young translations along the Pacific Coast.

Methods and Results

The Sonoma Volcanics is generally well exposed over much of its areal extent; however, structural attitudes cannot always be determined accurately, making outcrops suitable for paleomagnetic sampling difficult to find. Sedimentary bedding was used to determine structural attitude wherever available because the use of flow tops for this purpose can introduce considerable error by indicating initial dip rather than tectonic dip. During the current study, 115 cores were obtained from 14 sites using a gasoline-powered portable core drill. All but one of these sites were in the lower member of the Sonoma Volcanics that Fox et al. [1985*b*] partially subdivide informally as the andesites of Rodgers Creek and Atlas Peak. These two map units range in age from about 8 Ma to about 5.5 Ma and from about 5.5 Ma to about

TABLE 1. Paleomagnetic Data for the Sonoma Volcanics, California

Map number	Site Lat.	Site Long.	N	H	Uncorrected Incl.	Uncorrected Decl.	Corrected Incl.	Corrected Decl.	k	α_{95}	J_{nrm}	J_{cl}	X	Q	Corrected VGP Lat.	Corrected VGP Long.
Tuff of Petrified Forest (part of upper member of *Fox et al.* [1985*b*])																
SV27 (S28)	38.66	237.38	5	400	50.8	0.0	50.8	0.0	93	8.0	n.d.	1.11	n.d.	n.d.	82.8	57.4
SV26	38.56	237.37	7	200	-49.3	177.3	-51.4	164.3	8300	0.7	10.2	10.5	8.33	2	-75.6	305.1
SV25 (S7)	38.56	237.53	8	200	-50.0	181.3	-68.2	166.4	906	1.9	6.84	7.66	9.93	1	-74.1	25.1
SV24 (S6)	38.56	237.53	8	50	-11.8	189.0	-31.3	187.4	518	2.4	4.99	5.08	1.32	7	-67.4	218.9
Andesite of Atlas Peak (part of lower member of *Fox et al.* [1985*b*])																
SV23	38.36	237.79	8	400	32.0	37.9	32.0	37.9	1270	1.6	4.49	1.01	5.63	2	50.9	349.5
SV22	38.37	237.74	8	100	42.1	354.2	50.9	352.3	729	2.1	9.69	7.86	5.87	3	80.8	103.2
SV21	38.42	237.70	11	100	41.1	326.7	29.2	333.2	204	3.2	8.76	9.13	3.45	5	57.2	111.0
SV20	38.41	237.76	8	100	-51.3	189.9	-41.3	213.4	1688	1.4	4.07	4.05	0.86	9	-58.1	165.3
SV19(S4,S5)	38.35	237.77	16	200	-61.1	191.4	-47.6	181.9	142	3.1	0.69	1.46	1.97	1	-80.3	227.8
SV18 (S3)	38.35	237.77	8	200	-66.9	203.9	-54.6	188.1	179	4.2	14.4	8.97	8.51	3	-82.8	171.9
SV17 (S2)	38.35	237.77	8	50	-49.3	186.0	-35.5	180.6	401	2.8	0.66	0.66	0.45	3	-71.3	236.1
SV16	38.30	237.79	8	100	-55.1	200.6	-49.1	185.6	995	1.8	24.5	23.6	25.5	2	-80.5	207.0
SV15	38.30	237.79	8	400	-40.3	185.0	-32.2	177.6	86	6.0	n.d.	0.64	n.d.	n.d.	-69.1	244.3
SV14	38.30	237.79	8	200	-57.9	172.6	-47.5	162.2	390	2.8	20.5	8.01	15.8	3	-72.3	300.0
SV13	38.29	237.78	8	400	-51.1	194.2	-44.2	182.0	1192	1.6	63.0	27.5	9.65	13	-77.5	229.5
SV12	38.29	237.78	8	200	-50.9	198.2	-44.7	185.5	853	1.9	20.7	20.6	22.0	2	-77.2	215.1
SV11 (S19)	38.43	237.45	9	50	49.9	359.6	62.5	335.6	167	4.0	52.1	45.1	6.42	16	70.9	171.6
SV10 (S16)	38.47	237.38	8	50	-15.8	187.2	-27.1	176.0	543	2.4	10.2	10.2	3.35	6	-65.6	246.8
SV9	38.47	237.38	7	300	-17.3	165.5	-38.8	150.7	292	3.5	n.d.	3.25	n.d.	n.d.	-60.0	302.4
SV8 (S15)	38.47	237.38	7	200	-20.9	153.2	-40.2	131.2	94	6.3	11.5	4.26	15.4	2	-45.8	321.4
SV7 (S14)	38.48	237.38	6	100	6.0	1.8	47.7	345.8	498	3.0	11.5	11.1	1.97	11	74.7	112.0
SV6 (S12)	38.48	237.39	8	100	9.1	182.7	-40.5	180.4	343	3.0	55.7	47.6	3.06	35	-74.6	235.9
SV5 (S11)	38.48	237.39	8	200	38.5	200.0	-40.6	200.3	80	6.2	n.d.	4.34	n.d.	n.d.	-66.9	182.9
Andesite of Rodgers Creek (part of lower member of *Fox et al.* [1985*b*])																
SV4 (S21)	38.47	237.3	8	100	-45.1	161.8	-45.1	161.8	42	8.6	2.06	2.15	2.66	2	-70.7	294.9
SV3	38.50	237.27	8	300	-61.1	241.4	-65.5	203.6	262	3.4	31.4	13.1	7.36	8	-70.6	111.7
SV2	38.30	237.58	8	300	-4.8	356.1	-4.8	356.1	236	3.6	n.d.	0.69	n.d.	n.d.	49.2	63.5
SV1	38.31	237.55	7	100	50.0	330.2	50.0	330.2	117	5.6	21.5	3.62	10.4	4	64.5	139.4

Lat., Long.= north latitude and east longitude, respectively; N= number of samples; H= demagnetizing field (Oe); Incl.= inclination, positive downwards; Decl.= declination, east of north; k= precision parameter, $(N-1)/(N-R)$; α_{95}= 95% confidence cone about average direction; J_{nrm}, J_{cl}= geometric mean intensity before and after magnetic cleaning, respectively ($\times 10^{-4}$ emu/cm^3); X= geometric mean susceptibility ($\times 10^{-4}$); Q= geometric mean in-situ ratio of remanent to induced magnetization (Koenigsberger ratio); VGP= virtual geomagnetic pole; n.d.= not determined. Numbers within parentheses are flow designations used by Mankinen [1972].

4 Ma, respectively. The remaining site was in the rhyolite of Calistoga (part of the upper member of the Sonoma Volcanics [Fox et al., 1985*b*]) which ranges from about 3 Ma to about 2.6 Ma. Cores were oriented using a sun compass, and remanent magnetization was measured using a cryogenic magnetometer. Stability of remanence was tested by progressive alternating-field (AF) demagnetization of at least one specimen from each lava flow. One specimen from each of the remaining cores in a flow was then cleaned at a field strength that was chosen on the basis of behavior during the progressive demagnetization experiments.

Magnetically cleaned remanence directions before and after correcting for tilt are given in Table 1. Flows whose data were used by Mankinen [1972] to calculate a mean direction are also listed in the table, and all sites are shown in Figure 1. Sites S4 and S5 from the earlier study appear to belong to different parts of the same welded tuff, and data from these sites are combined. Cores were obtained from hand-sample locality S20 and this site is now designated as SV3 in Table 1. The precision parameter, k, for the sample population increases from 8 to 17 when the data are structurally corrected. This fold test is significant at the 99% confidence level [McElhinny, 1964]. The mean directions calculated for the corrected normal- (I= 42.2°, D= 352.5°, k= 9) and reversed- (I= -46.0°, D= 177.7°, k= 23) polarity data differ by 174.7° and are antipodal at the 95% confidence level [McFadden and Jones, 1981]. In a study of the middle Miocene Steens Basalt, Mankinen et al. [1985] concluded that a virtual geomagnetic pole (VGP) latitude of less than 50° defined the transitional geomagnetic field in southeast Oregon. The same limit is assumed to be applicable for these data, and results for flows SV2 and SV8 (Table 1) are not used in the calculation of a mean paleopole. The remaining 25 flows yield a paleopole at 80.2°N., 069.2°E., with k= 19 and α_{95}= 6.8°. The total angular dispersion, S_T, of VGP's about the paleopole is 18.6°, and the dispersion corrected for within-site dispersion due to experimental errors, S_F, is 18.5°. The upper and lower limits of 95% confidence for S_F are equal to 23.1° and 15.4°, respectively [Cox, 1969]. Using all the data in Table 1 yields a paleopole at 78.8°N., 076.2°E., with α_{95}= 7.2° and S_F= 20.5°. Neither paleopole differs statistically from that previously calculated (79.4°N., 083.3°E., α_{95}= 8.9° [Mankinen, 1972]).

Discussion and Conclusions

In many respects, this seems to be a reasonable data set. Data from 25 cooling units were used to calculate the paleomagnetic pole, and the possibility of multiple

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sampling of a given geomagnetic field direction seems to be minimal. Where a number of flows were sampled at a given locality, such as Los Alamos Road (SV5 to SV10), each has a remanence direction that is distinct from other flows in the sequence. A normal-polarity lava flow, probably representing a magnetic-polarity subchron, is bracketed between several reversed-polarity flows at this latter locality [Mankinen, 1972]. Two reversed-polarity welded tuffs along Monticello Road yield K-Ar ages [Mankinen, 1972] of 4.8 ± 0.2 Ma (SV17) and 4.3 ± 0.1 Ma (SV19) when corrected to conform with currently accepted abundance and decay constants. A normal-polarity welded tuff (SV23) occurs near the top of this sequence. The lowermost two andesite flows along Wild Horse Canyon Road (SV12 to SV16) have similar directions and may have erupted at nearly the same time, but the directions recorded by the remaining flows are distinct. K-Ar ages of 5.02 ± 0.22 Ma and 3.4 ± 0.2 Ma [Fox et al., 1985*b*] have been obtained for flows within this sequence, but Fox et al. believe that the 3.4-Ma age is too young on the basis of geologic correlations with other dated units. Statistically different magnetization directions and radiometric ages, as well as numerous interbeds of tuff and tuffaceous sediment, all indicate appreciable time intervals between successive eruptions. Some duplicate recording of a geomagnetic field direction could have occurred, however, if vents in different areas of the volcanic field were active at the same time.

The data set is weighted heavily toward reversed polarities, but most of the units sampled were erupted between about 5.5 Ma and 4 Ma when the geomagnetic field was predominately reversed (the Gilbert Reversed-Polarity Chron). The total time span represented by the data in Table 1 is from about 8 Ma to about 3 Ma. Flow directions generally are well determined (Table 1), and the data yield positive fold and reversal tests. Moreover, the ancient geomagnetic field dispersion for the Sonoma Volcanics (18.5°) is virtually identical to that calculated by Mankinen et al. [1987] for the Steens Basalt (18.2°) and similar to that for the Columbia River Basalt Group (16.9°). This dispersion also is somewhat, but not significantly, larger than the 15.5° (confidence limits of 16.6° and 14.5°) calculated for the same latitude range for the past 5 million years [McFadden et al., 1988]. Despite these positive attributes, however, the revised paleopole differs significantly from the geographic axis (Figure 2) by the same amount and in the same direction as did the pole calculated in the previous study.

In order to test for tectonic rotation and/or translation, a reference pole of the appropriate age must be determined. Mankinen et al. [1987] have shown that the paleopoles for the Steens Basalt and the Columbia River Basalt Group are both within 3° of, and statistically indistinguishable from, the rotational axis. Because the Sonoma Volcanics is younger than either of those units, the rotational axis is clearly the best reference to use here. A comparison using this reference shows that the Sonoma Volcanics paleopole is displaced 9.6° ± 5.3° of latitude to the farside of the geographic pole (i.e., the average magnetic inclination is too shallow by 11.2°) and is not significantly displaced from the site meridian (2.2° ± 6.6° westward). Note that the uncertainties cited are due entirely to the dispersion of the Sonoma Volcanics data set and would be greater if the comparison were made with a paleomagnetically derived reference pole.

The magnetic anisotropy (K_{max}/K_{min}) of representative samples from these volcanics ranges from 1.01 to 1.06, and is too small to cause significant deflection of the remanence direction [e.g., McElhinny, 1973]. Because the data are compared with the geographic axis, however, the inclination anomaly could be due, in part, to non-

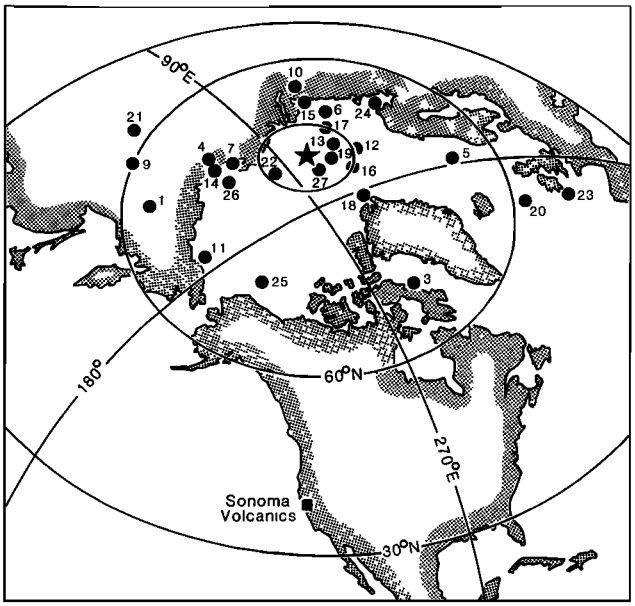


Fig. 2. Virtual geomagnetic pole positions (dots) for reliable volcanic units from the Sonoma Volcanics with reversed-polarity directions inverted through the origin. Star is mean paleopole with its associated 95% confidence limit. Numbers refer to Table 1.

zonal harmonics of the geomagnetic field. Zonal asymmetry presumably is included in paleomagnetic data used to calculate reference poles for the craton and need not be considered when a comparison is made with any of these. The expected difference in inclination, ΔI, between the observed field and that due to a reversed geocentric axial dipole in the northern hemisphere at the latitude of the Sonoma Volcanics is on the order of about -5° on the basis of an analysis of world-wide paleomagnetic data from igneous rocks younger than 5 Ma [Lee, 1983]. An analysis based on inclination-only data from equatorial deep-sea cores [Schneider and Kent, 1988] indicates a slightly smaller ΔI (-4°) for the area during the Gilbert Chron. Thus, only about one-half of the inclination anomaly indicated by this data set can be reasonably attributed to nonzonal harmonics.

Luyendyk et al. [1985] noted that most paleomagnetic studies of Neogene volcanic and sedimentary rocks from southern and Baja California, and southwest Arizona yield magnetic inclinations that are shallower than expected when compared with cratonic reference poles. Because many of the areas sampled are in regions clearly attached to the craton, Luyendyk et al. postulated that the inclination anomalies could be caused by long-period nondipole geomagnetic field behavior. Some of the studies, however, are based on limited sampling and are unlikely to have averaged secular variation; others may have used improper structural corrections (see summaries in Luyendyk et al. [1985] and Hagstrum et al. [1987]). A paleopole calculated from 64 volcanic units in Baja California [Hagstrum et al., 1987] does not differ significantly from the reference pole for the craton, a result that dispels the possibility of a persistent nondipole field in the region during Miocene time. Such a field, therefore, is not a viable explanation for the shallow inclination recorded by the Sonoma Volcanics because the two areas are separated only by 10° of latitude and 7° of longitude.

The Sonoma Volcanics VGP's do not appear to be circularly symmetric (Figure 2) and Engebretson and Beck [1978] statistics show that the distribution is, indeed, highly elliptical (axial ratio = 1.69). The plane of elon-

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gation (pole at 9.7°S., 62.2°E.) is nearly perpendicular to the site meridian, and the axial ratio is larger than that introduced by the VGP mapping function (ratio = 1.36) to a symmetrical set of directions having inclinations typical for this latitude (see Hagstrum et al. [1987] for further discussion). The site-mean directions also have an elongate distribution (axial ratio = 1.23) that is nearly perpendicular to the site meridian (pole at I= 42.9°, D= 166.6°). Although fully representative data are expected to produce a circular distribution in either direction or VGP space (e.g., Merrill and McElhinny [1983]), effects of random errors, a random nondipole field, and dipole wobble can ensure that neither poles nor directions are entirely circular. Even in such cases, however, an elongation perpendicular to the site meridian in one space should produce an elongation along the meridian in the other because of the mapping function. The Sonoma Volcanics data set is anomalous in that both directions and poles are elongate perpendicular to the site meridian. Because the excess dispersion seems to be in declination, undetected vertical-axis rotations, incorrect structural corrections, and orientation errors could be contributing factors. Declination anomalies, however, cannot be ascribed to any specific area within the volcanic field, and random rotations of individual lava flows should have a minimal effect on the overall population statistics. Vertical-axis rotation also could not produce the shallow inclination. Using incorrect structural attitudes to correct for tilt could cause inclination and/or declination errors, but the fact that these data pass a fold test at greater than 99% confidence makes it extremely unlikely that significant errors have been introduced in this manner. Cores were oriented either with a sun compass or, in the earlier study, a magnetic compass utilizing back azimuths for corrections, which also rules out field procedures as a probable source of error. The most likely explanation for the elliptical distribution of the data set and the anomalous inclination measured in the Sonoma Volcanics, therefore, is that the effects of secular variation have not been averaged. This conclusion points out the difficulty in determining when secular variation has been averaged in any given volcanic field and indicates that the total number of sites alone can be a poor measure.

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